

1ai	Taking moments about X, $(3.0)(0.30)(12) = (2)(0.30)(d)$ $d = 18 \text{ cm}$
1aii	$k_{\text{eff}} = k_1 + k_2$ $= 3.0 + 2.0$ $= 5.0 \text{ N m}^{-1}$ Or Let the effective spring constant be k_{eff} $k_{\text{eff}} x = mg$ $k_{\text{eff}} (0.30) = 0.15(10)$ $k_{\text{eff}} = 5.0 \text{ N m}^{-1}$
1bi	Let the new extension be e. $3e + 2e = (0.15 + 0.050)(10)$ $e = 0.40 \text{ m (or 40 cm)}$
1bii	Difference in EPE $= \frac{1}{2} (3.0+2.0)(0.40)^2 - \frac{1}{2} (3.0+2.0)(0.30)^2$ $= 0.175 \text{ J}$ Alternatively, difference in EPE $= \frac{1}{2} F_2 x_2 - \frac{1}{2} F_1 x_1$ $= \frac{1}{2} (0.20)(10)(0.4) - \frac{1}{2} (0.15)(10)(0.3)$ $= 0.175 \text{ J}$
1c	Let the new CG be y cm away from X. Taking moments about G (the new CG), $(0.15)(10)(y) = (0.050)(10)(8-y)$ $y = 2.00 \text{ cm}$

2ai The gravitational potential at a point is the work done per unit mass (by an external force)